Calc 3 – Multiple Integrals

Worksheet A

1. Evaluate $\int_0^1 \int_0^2 (x + 2y)\, dy\, dx$.

2. Sketch and evaluate $\iint_D xy\, dA$ where D is bounded by $y = x$ and $y = x^2$.

3. Convert to polar and evaluate $\iint_D e^{-(x^2+y^2)}\, dA$ where D is the disk $x^2 + y^2 \leq 1$.

4. Evaluate $\iiint_E z\, dV$ where $E = [0, 1]^3$.

5. Use cylindrical coordinates to find the volume of the solid bounded above by $z = 4 - x^2 - y^2$ and below by $z = 0$.

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Worksheet B

1. Evaluate $\int_0^1 \int_{x^2}^{x} (x^2 + y^2)\, dy\, dx$.

2. Reverse the order of integration: $\int_0^1 \int_y^1 \sin(x^2)\, dx\, dy$, then evaluate.

3. Convert to polar and evaluate $\iint_D x^2\, dA$ where D is the upper half-disk $x^2 + y^2 \leq 4, y \geq 0$.

4. Use spherical coordinates to compute the volume of the ball $x^2 + y^2 + z^2 \leq 4$.

5. Find the volume of the solid in the first octant bounded by the plane $x + y + z = 1$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. Inner: $\int_0^2 (x+2y)\,dy = 2x+4$. Outer: $\int_0^1 (2x+4)\,dx = 1+4 = \boxed{5}$.

2. D: $0 \le x \le 1$, $x^2 \le y \le x$. $\iint xy\,dA = \int_0^1 x \int_{x^2}^x y\,dy\,dx = \int_0^1 x \cdot \frac{x^2-x^4}{2}\,dx = \frac{1}{2}\int_0^1 (x^3 - x^5)\,dx = \frac{1}{2}(1/4 - 1/6) = \boxed{1/24}$.

3. $\iint e^{-r^2} r\,dr\,d\theta = \int_0^{2\pi}\int_0^1 re^{-r^2}\,dr\,d\theta = 2\pi \cdot [-\frac{1}{2}e^{-r^2}]_0^1 = \boxed{\pi(1-e^{-1})}$.

4. $\int_0^1\int_0^1\int_0^1 z\,dz\,dy\,dx = \int_0^1 \frac{1}{2}\,dy\,dx = \boxed{1/2}$.

5. $V = \int_0^{2\pi}\int_0^2\int_0^{4-r^2} r\,dz\,dr\,d\theta = 2\pi\int_0^2 r(4-r^2)\,dr = 2\pi \cdot [2r^2 - r^4/4]_0^2 = 2\pi(8-4) = \boxed{8\pi}$.

Worksheet B.

1. Inner: $\int_{x^2}^x (x^2+y^2)\,dy = x^2(x-x^2) + \frac{x^3-x^6}{3} = x^3 - x^4 + \frac{x^3-x^6}{3} = \frac{4x^3-3x^4-x^6}{3}$. Integrate: $\int_0^1 = \frac{1}{3}(1 - 3/5 - 1/7) = \frac{1}{3} \cdot \frac{35-21-5}{35} = \frac{9}{105} = \boxed{3/35}$.

2. Region: $0 \le y \le x \le 1$. Swap: $\int_0^1\int_0^x \sin(x^2)\,dy\,dx = \int_0^1 x\sin(x^2)\,dx = \boxed{\frac{1}{2}(1-\cos 1)}$.

3. $x^2 = r^2\cos^2\theta$, $dA = r\,dr\,d\theta$. $\int_0^\pi \cos^2\theta\,d\theta \int_0^2 r^3\,dr = \frac{\pi}{2} \cdot 4 = \boxed{2\pi}$.

4. $V = \int_0^{2\pi}\int_0^\pi\int_0^2 \rho^2 \sin\varphi\,d\rho\,d\varphi\,d\theta = 2\pi \cdot 2 \cdot \frac{8}{3} = \boxed{\frac{32\pi}{3}}$.

5. $V = \int_0^1\int_0^{1-x}\int_0^{1-x-y} 1\,dz\,dy\,dx = \int_0^1\int_0^{1-x} (1-x-y)\,dy\,dx = \int_0^1 \frac{(1-x)^2}{2}\,dx = \boxed{1/6}$.